

Status — main arm (PH vs classical)

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1. What changed

Cloud bank (replaces the $\tilde{\delta}$ -sweep)

- 50 000 $\theta \times 2$ patterns $\times 5$ families = 500k clouds (was 10k θ); $\bar{n} \sim \log\text{-U}[75, 950]$
- Drawn on *physical* parameters, two independent dials: richness (points/cluster) and smearing $\omega = \sigma\sqrt{\kappa} \in [0.2, 8]$
- $\tilde{\delta}$ is now a post-hoc *label*: refitting the null tables relabels, never re-simulates
- Bounds inverted into the draw's support, not rejected: 0 rejections (LGCP 1.7%, numerical grid limit). Old design: nested rejected 18% and tilted the prior
- $\text{cv}(n) \leq 0.30$ (was 0.45): the label is evaluated at $\bar{n}$ but describes realised n ; drift $\approx$ linear in cv
- Test set: 12 000 θ /family (was 2 000), so near-CSR cells are populated

Regime analysis

- $\tilde{\delta}$ reduction **sup**→**ext**: extremum of $\hat{L} - r$ *before* studentising. No radius window, exact size (0.0501). Fixes Matérn hard cores below $r_{\min}$ (power 19% $\rightarrow$ 81% at $\bar{n} = 200$)
- Unclamped: $\tilde{\delta} = 0$ is CSR's noise floor, Poisson sits at -0.97 ; $\tilde{\delta} = 1$ is the 5% rejection boundary
- Signed arms $\tilde{\delta}_{\text{lo}}$ (repulsion), $\tilde{\delta}_{\text{hi}}$ (clustering) stored as alternative coordinates
- 10 geometric $\tilde{\delta}$ bins on $[0.05, 32] \times 3 \bar{n}$ bins; bootstrap over test θ ; paired differences
- Undetectable share ($\tilde{\delta} < 1$): Thomas 55%, LGCP 52%, Matérn 29%, nested 20%
- Below the first bin (unbinned, marginal only): Thomas 42%, LGCP 39%, nested 13%, Matérn 10%

PersLay (third arm)

- Same diagrams, learned vectorisation $V(D) = \text{op}\{w(p)\varphi(p)\}$, $Q = 64$, sum pooling, $\approx 4.4\text{k}$ parameters per dim
- Variants: plain, and **z** (pooled vector z-scored)
- Run: classification only (plain 10 seeds, **z** 2 seeds so far). **Not yet run for parameters**

2. Classification (5-way, chance 0.2)

	overall	Pois.	Thomas	nested	Matérn	LGCP
L, F, G, J fixed	0.604	0.812	0.160	0.611	0.882	0.552
L fixed (VIHRS)	0.583	0.817	0.150	0.571	0.890	0.487
PersLay-z DTM ₁₀ H_{0+1}	0.585	0.799	0.147	0.596	0.845	0.536
PersLay DTM ₁₀ H_{0+1}	0.577	0.830	0.130	0.600	0.853	0.470
image alpha H_{0+1} (best image)	0.563	0.828	0.119	0.562	0.771	0.533
image DTM ₁₀ H_{0+1}	0.549	0.802	0.144	0.585	0.674	0.539

Accuracy by $\tilde{\delta}$ (4 families; chance 0.25):

$\tilde{\delta}$ bin	0.05	0.10	0.18	0.35	0.66	1.26	2.41	4.61	8.79	16.8
L, F, G, J fixed	.279	.338	.379	.485	.595	.626	.684	.773	.819	.831
PersLay-z DTM ₁₀ H_{0+1}	.256	.309	.332	.404	.517	.605	.672	.755	.806	.823
image alpha H_{0+1}	.195	.227	.249	.304	.418	.563	.651	.739	.794	.817
image DTM ₁₀ H_{0+1}	.201	.225	.234	.270	.335	.491	.652	.758	.809	.828

- Classical still best overall; PersLay is now the best PH vectorisation
- PersLay-z halves the PH-classical gap: paired vs L, F, G, J **sqrtn**, image PH -0.033 to $-0.050 \rightarrow$ PersLay-z -0.012 $[-0.013, -0.010]$
- The gain is near CSR ($\tilde{\delta} < 1$): image PH is at/below chance there, PersLay-z recovers half or more of the gap. Above $\tilde{\delta} \approx 2$ all arms converge

- Main source: Matérn recall, DTM₁₀ 0.674 → 0.845 (image→PersLay)
- Thomas is unsolved by every arm (recall ≤ 0.17), and its recall is non-monotone in $\tilde{\delta}$: peaks at ≈ 0.34 (classical) at $\tilde{\delta} \approx 5$, drops to 0.12 at the top bin
- H_1 alone is the weakest input everywhere (≤ 0.50)
- Accuracy now *rises* with $\bar{n}$ (0.53 → 0.68): in the bank, more points means more information. The old sweep held $\tilde{\delta}$ fixed across $\bar{n}$ and showed the opposite

3. Parameter estimation (image PH vs classical)

Log-RMSE pooled over targets:

	Poisson	Thomas	nested	Matérn	LGCP
L, F, G, J fixed	0.067	0.754	0.571	0.111	0.535
L fixed (VIHRS)	0.066	0.757	0.576	0.122	0.535
DTM ₅ H_{0+1}	0.067	0.777	0.595	0.151	0.563
alpha H_{0+1}	0.067	0.795	0.611	0.104	0.569
Rips H_0	0.067	0.846	0.648	0.103	0.595

Paired PH – classical (L, F, G, J fixed) by $\tilde{\delta}$; positive = PH worse, * = 95% interval excludes 0:

		0.05	0.10	0.18	0.35	0.66	1.26	2.41	4.61	8.79	16.8
Matérn	alpha H_{0+1}	+0.004	+0.002	+0.004*	+0.003*	.000	−.002*	−.010*	−.030*	−.029*	−.039*
Thomas	DTM ₅ H_{0+1}	−.004	+0.013	+0.009	+0.015*	+0.041*	+0.049*	+0.052*	+0.030*	+0.054*	+0.005
nested	DTM ₅ H_{0+1}	+0.005	+0.010	+0.003	+0.004	+0.007*	+0.023*	+0.037*	+0.027*	+0.037*	+0.026*
LGCP	DTM ₅ H_{0+1}	+0.049*	+0.050*	+0.033*	+0.050*	+0.062*	+0.054*	+0.040*	+0.016*	+0.008*	−.017*

- Classical leads on Thomas, nested, LGCP; Poisson is a tie for every input (log n solves it)
- **Matérn is the PH win**, now sharper: PH ahead from $\tilde{\delta} \approx 1.3$, by 0.03–0.04 above $\tilde{\delta} \approx 5$, where the classical error stops falling (and L alone rises, 0.081 → 0.171)
- LGCP: PH worse near CSR, gap closes with $\tilde{\delta}$ and flips in the top bin
- Thomas/nested: tie near CSR, PH gap opens once the structure is detectable ($\tilde{\delta} \gtrsim 0.7$)
- By $\bar{n}$: Matérn and LGCP improve with $\bar{n}$; Thomas and nested get *harder* (0.66 → 0.85 Thomas, classical)

4. Next

- PersLay for parameter estimation (does it extend the Matérn win, close the LGCP gap?)
- Finish PersLay-z seeds (2/10)
- Decide how to present the sub-noise-floor region ($\tilde{\delta} < 0.05$ is $\approx 40\%$ of Thomas and LGCP)
- Diagnose Thomas: where its patterns go, and why recall falls at high $\tilde{\delta}$
- Predictive checks (distance to true process, envelope test) are implemented but not yet run on the new bank